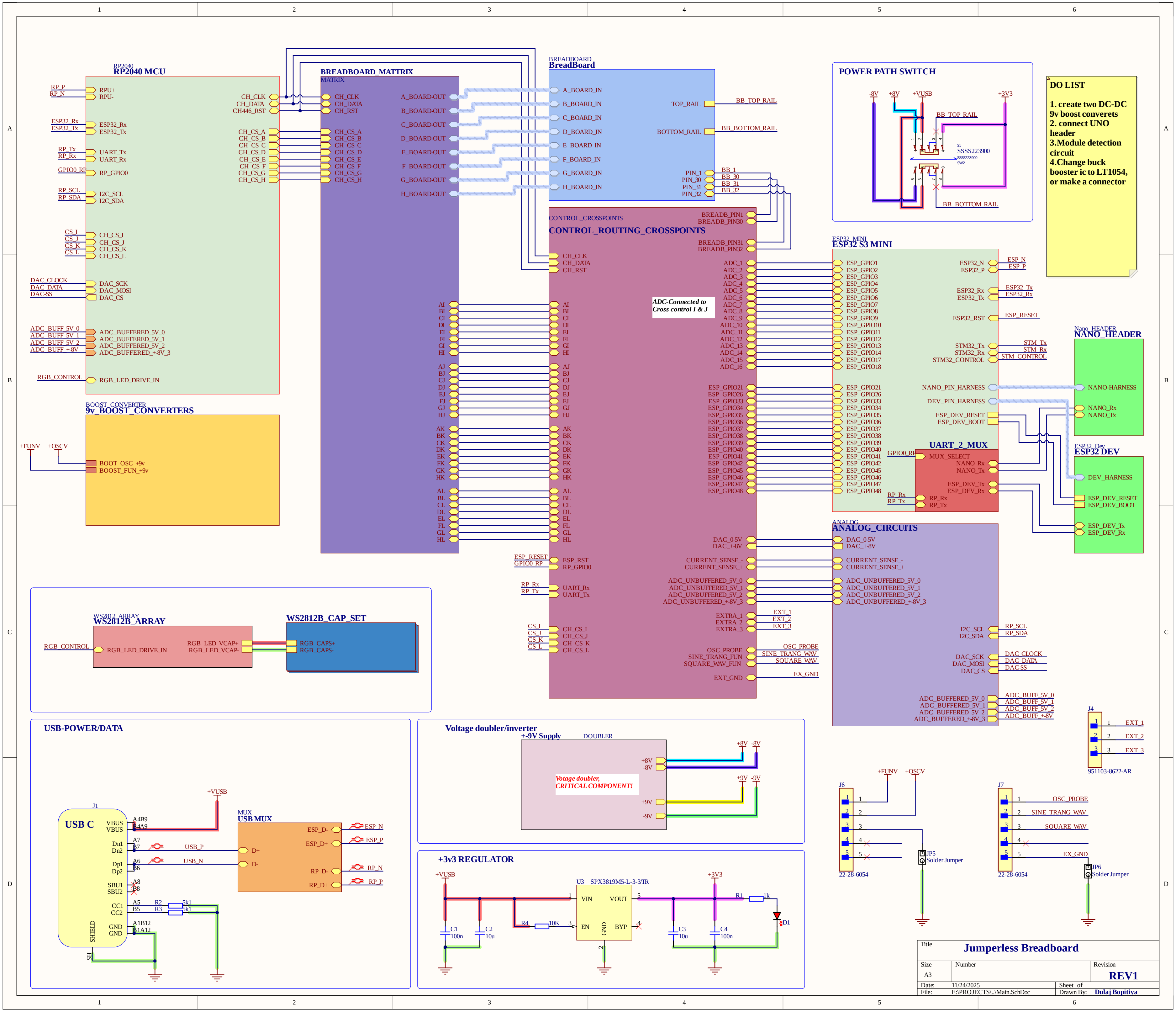
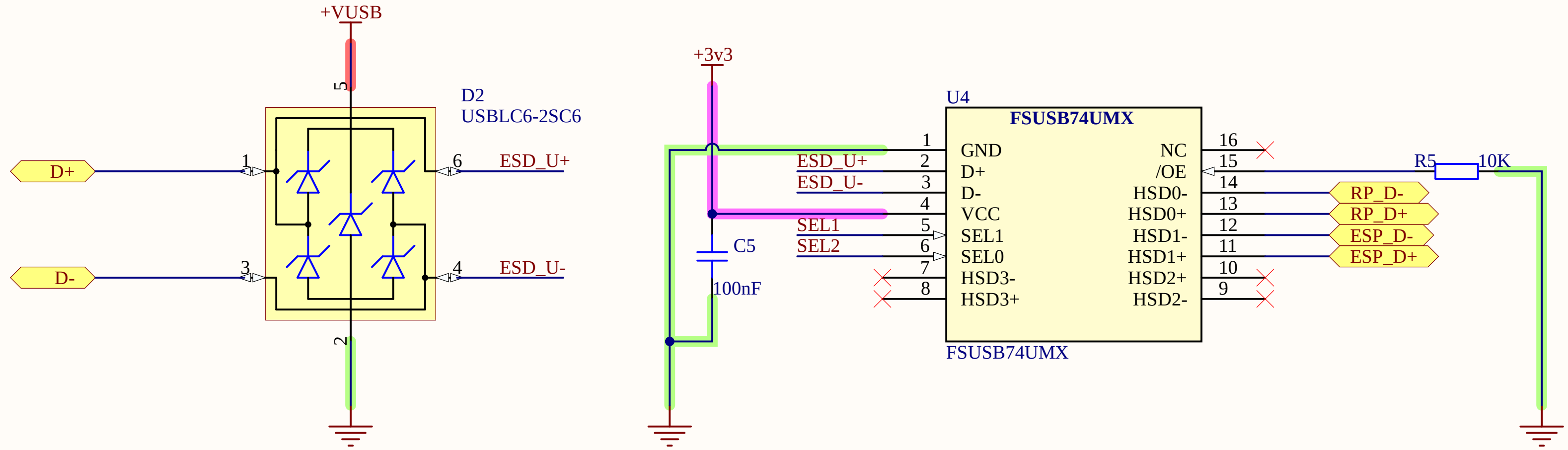
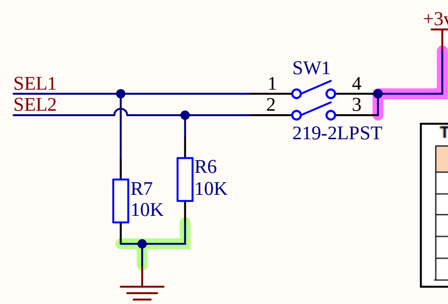


RP2040 MCU
BREADBOARD_MATTRIX
BreadBoard
POWER PATH SWITCH
DO LIST
1. create two DC-DC 9v boost converets
2. connect UNO header
3.Module detection circuit
4.Change buck booster ic to LT1054, or make a connector
CONTROL_ROUTING_CROSSPOINTS
ADC-Connected to Cross control I & J
ESP32 S3 MINI
NANO_HEADER
UART_2_MUX
ESP32 DEV
9v_BOOST_CONVERTERS
ANALOG_CIRCUITS
WS2812B_ARRAY
WS2812B_CAP_SET
USB-POWER/DATA
USB C
USB MUX
Voltage doubler/inverter
+-9V Supply
Votage doubler, CRITICAL COMPONENT!
+3v3 REGULATOR
SPX3819M5-L-3-3/TR
Title
Jumperless Breadboard
Size
A3
Number
Revision
REV1
Date: 11/24/2025
File: E:\PROJECTS\..\Main.SchDoc
Sheet of
Drawn By: Dulaj Bopitiya

ESD PROTECTION + USB MUX



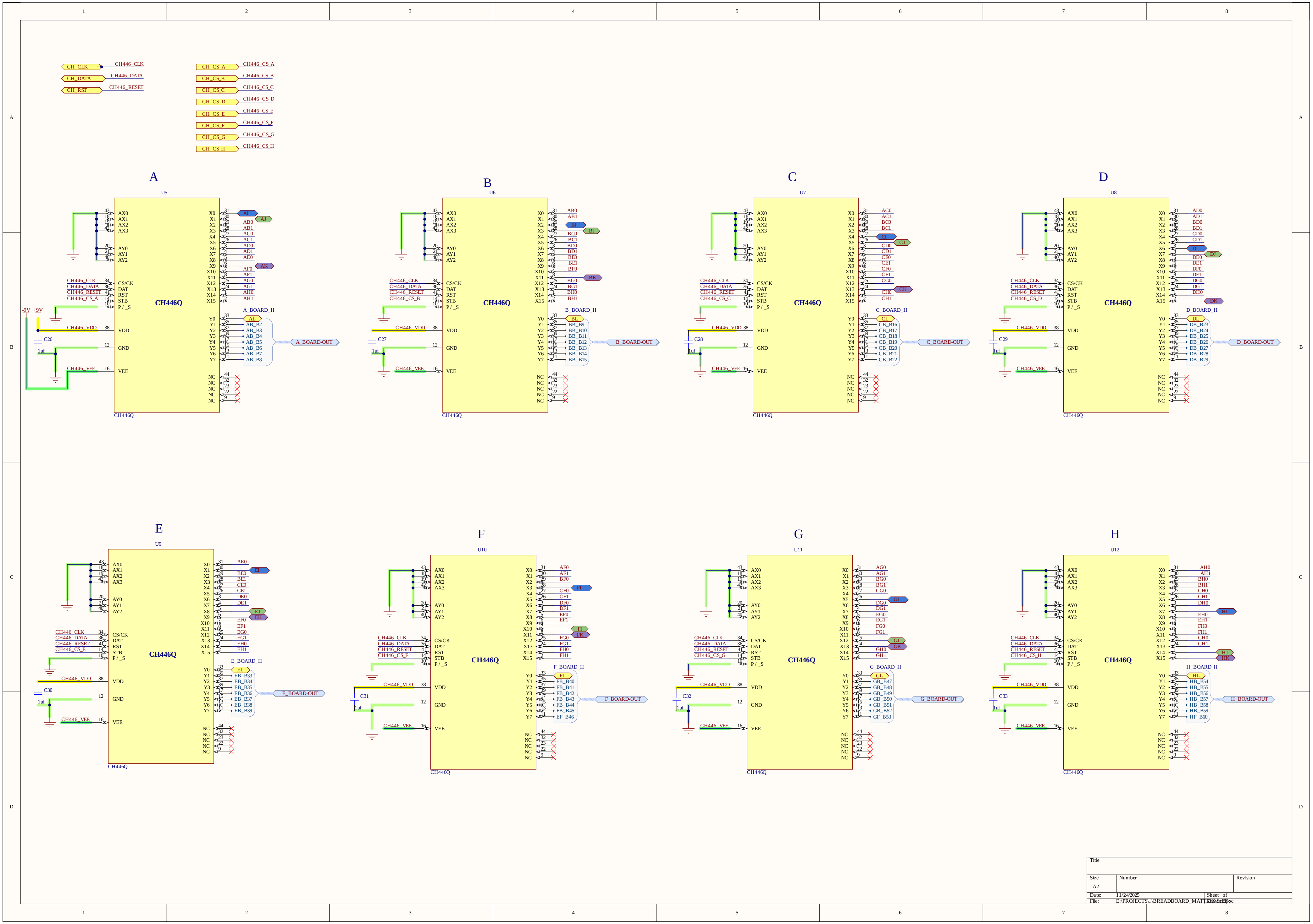
MUX SELECT SWITCH

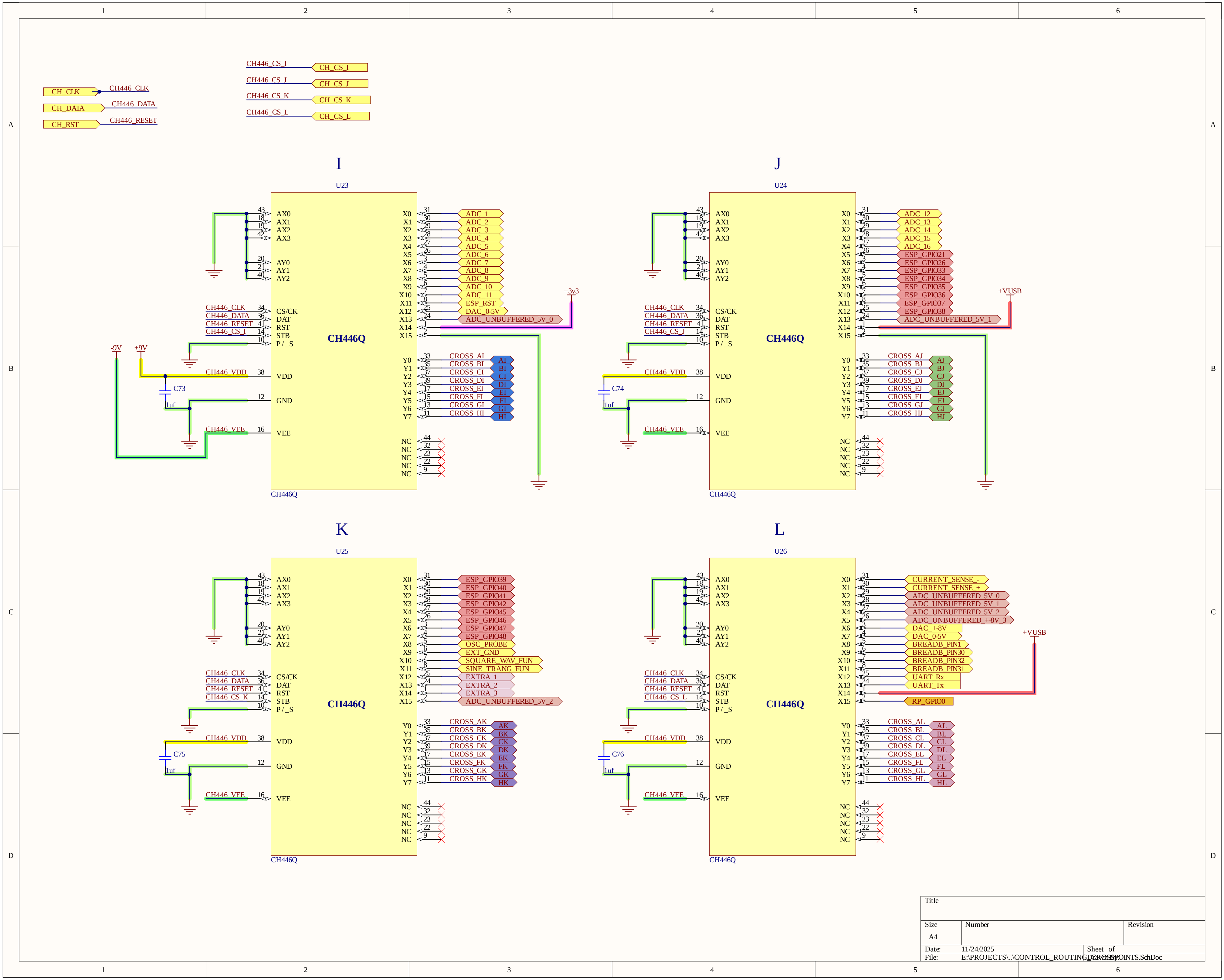


Truth Table

/OE	SEL1	SEL0	Function
1	X	X	D+, D- Switch Paths Open
0	0	0	D+=HSD0+, D-=HSD0-
0	0	1	D+=HSD1+, D-=HSD1-
0	1	0	D+=HSD2+, D-=HSD2-
0	1	1	D+=HSD3+, D-=HSD3-

Title	USB_MUX	
Size: A	Number	Revision: REV1
Date: 11/24/2025	Sheet of	
File: E:\PROJECTS\..\USB MUX.SchDoc	Drawn By: Dulaj Bopitiya	


CH_CLK CH446_CLK
CH_DATA CH446_DATA
CH_RST CH446_RESET
CH_CS_A CH446_CS_A
CH_CS_B CH446_CS_B
CH_CS_C CH446_CS_C
CH_CS_D CH446_CS_D
CH_CS_E CH446_CS_E
CH_CS_F CH446_CS_F
CH_CS_G CH446_CS_G
CH_CS_H CH446_CS_H
A
U5
CH446Q
A_BOARD_H
A_BOARD-OUT
C26
1uf
-9V
+9V
B
U6
B_BOARD_H
B_BOARD-OUT
C27
C
U7
C_BOARD_H
C_BOARD-OUT
C28
D
U8
D_BOARD_H
D_BOARD-OUT
C29
E
U9
E_BOARD_H
E_BOARD-OUT
C30
F
U10
F_BOARD_H
F_BOARD-OUT
C31
G
U11
G_BOARD_H
G_BOARD-OUT
C32
H
U12
H_BOARD_H
H_BOARD-OUT
C33
AX0 AX1 AX2 AX3 AY0 AY1 AY2 CS/CK DAT RST STB P / _S VDD GND VEE
X0 X1 X2 X3 X4 X5 X6 X7 X8 X9 X10 X11 X12 X13 X14 X15
Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 NC
CH446_VDD
CH446_VEE
Title
Size
A2
Number
Revision
Date: 11/24/2025
File: E:\PROJECTS\..\BREADBOARD_MAT...
Sheet of


CH_CLK
CH446_CLK
CH_DATA
CH446_DATA
CH_RST
CH446_RESET
CH446_CS_I
CH_CS_I
CH446_CS_J
CH_CS_J
CH446_CS_K
CH_CS_K
CH446_CS_L
CH_CS_L
I
U23
J
U24
K
U25
L
U26
CH446Q
AX0
AX1
AX2
AX3
AY0
AY1
AY2
CS/CK
DAT
RST
STB
P / _S
VDD
GND
VEE
X0
X1
X2
X3
X4
X5
X6
X7
X8
X9
X10
X11
X12
X13
X14
X15
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
NC
ADC_1
ADC_2
ADC_3
ADC_4
ADC_5
ADC_6
ADC_7
ADC_8
ADC_9
ADC_10
ADC_11
ESP_RST
DAC_0-5V
ADC_UNBUFFERED_5V_0
+3v3
CROSS_AI
CROSS_BI
CROSS_CI
CROSS_DI
CROSS_EI
CROSS_FI
CROSS_GI
CROSS_HI
AI
BI
CI
DI
EI
FI
GI
HI
-9V
+9V
CH446_VDD
C73
1uf
CH446_VEE
ADC_12
ADC_13
ADC_14
ADC_15
ADC_16
ESP_GPIO21
ESP_GPIO26
ESP_GPIO33
ESP_GPIO34
ESP_GPIO35
ESP_GPIO36
ESP_GPIO37
ESP_GPIO38
ADC_UNBUFFERED_5V_1
+VUSB
CROSS_AJ
CROSS_BJ
CROSS_CJ
CROSS_DJ
CROSS_EJ
CROSS_FJ
CROSS_GJ
CROSS_HJ
AJ
BJ
CJ
DJ
EJ
FJ
GJ
HJ
C74
ESP_GPIO39
ESP_GPIO40
ESP_GPIO41
ESP_GPIO42
ESP_GPIO45
ESP_GPIO46
ESP_GPIO47
ESP_GPIO48
OSC_PROBE
EXT_GND
SQUARE_WAV_FUN
SINE_TRANG_FUN
EXTRA_1
EXTRA_2
EXTRA_3
ADC_UNBUFFERED_5V_2
CROSS_AK
CROSS_BK
CROSS_CK
CROSS_DK
CROSS_EK
CROSS_FK
CROSS_GK
CROSS_HK
AK
BK
CK
DK
EK
FK
GK
HK
C75
CURRENT_SENSE_-
CURRENT_SENSE_+
ADC_UNBUFFERED_5V_0
ADC_UNBUFFERED_5V_1
ADC_UNBUFFERED_5V_2
ADC_UNBUFFERED_+-8V_3
DAC_+-8V
DAC_0-5V
BREADB_PIN1
BREADB_PIN30
BREADB_PIN32
BREADB_PIN31
UART_Rx
UART_Tx
RP_GPIO0
CROSS_AL
CROSS_BL
CROSS_CL
CROSS_DL
CROSS_EL
CROSS_FL
CROSS_GL
CROSS_HL
AL
BL
CL
DL
EL
FL
GL
HL
C76
Title
Size
A4
Number
Revision
Date:
11/24/2025
Sheet of
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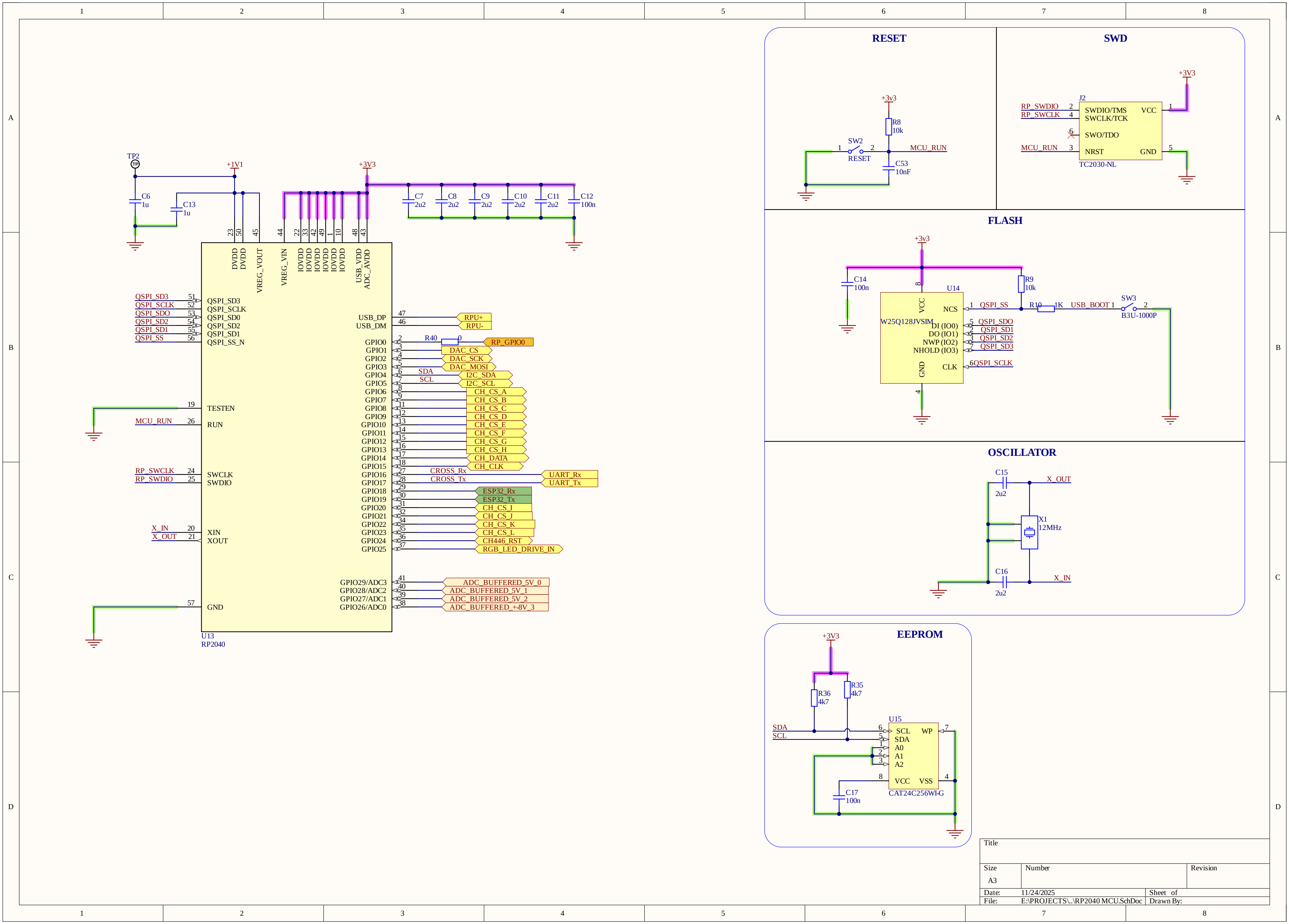
Drawn By:


RESET
+3v3
R8
10k
SW2
RESET
MCU_RUN
C53
10nF
SWD
+3V3
J2
RP_SWDIO
RP_SWCLK
MCU_RUN
SWDIO/TMS
SWCLK/TCK
SWO/TDO
NRST
VCC
GND
TC2030-NL
FLASH
+3v3
C14
100n
R9
10k
U14
W25Q128JVSIM
VCC
NCS
DI (IO0)
DO (IO1)
NWP (IO2)
NHOLD (IO3)
CLK
GND
QSPI_SS
QSPI_SDO
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
R10
1K
USB_BOOT
SW3
B3U-1000P
OSCILLATOR
C15
2u2
X_OUT
X1
12MHz
C16
2u2
X_IN
EEPROM
+3V3
R36
4k7
R35
4k7
U15
SDA
SCL
SCL
SDA
A0
A1
A2
WP
VCC
VSS
CAT24C256WI-G
C17
100n
TP2
C6
1u
C13
1u
+1V1
+3V3
C7
2u2
C8
2u2
C9
2u2
C10
2u2
C11
2u2
C12
100n
DVDD
DVDD
VREG_VOUT
VREG_VIN
IOVDD
IOVDD
IOVDD
IOVDD
IOVDD
IOVDD
USB_VDD
ADC_AVDD
QSPI_SD3
QSPI_SCLK
QSPI_SDO
QSPI_SD2
QSPI_SD1
QSPI_SS
QSPI_SD3
QSPI_SCLK
QSPI_SD0
QSPI_SD2
QSPI_SD1
QSPI_SS_N
TESTEN
MCU_RUN
RUN
RP_SWCLK
RP_SWDIO
SWCLK
SWDIO
X_IN
X_OUT
XIN
XOUT
GND
U13
RP2040
USB_DP
USB_DM
RPU+
RPU-
R40
0
RP_GPIO0
DAC_CS
DAC_SCK
DAC_MOSI
SDA
SCL
I2C_SDA
I2C_SCL
CH_CS_A
CH_CS_B
CH_CS_C
CH_CS_D
CH_CS_E
CH_CS_F
CH_CS_G
CH_CS_H
CH_DATA
CH_CLK
CROSS_Rx
CROSS_Tx
UART_Rx
UART_Tx
ESP32_Rx
ESP32_Tx
CH_CS_I
CH_CS_J
CH_CS_K
CH_CS_L
CH446_RST
RGB_LED_DRIVE_IN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO29/ADC3
GPIO28/ADC2
GPIO27/ADC1
GPIO26/ADC0
ADC_BUFFERED_5V_0
ADC_BUFFERED_5V_1
ADC_BUFFERED_5V_2
ADC_BUFFERED_+8V_3
Title
Size
A3
Number
Revision
Date: 11/24/2025
Sheet of
File: E:\PROJECTS\..\RP2040 MCU.SchDoc
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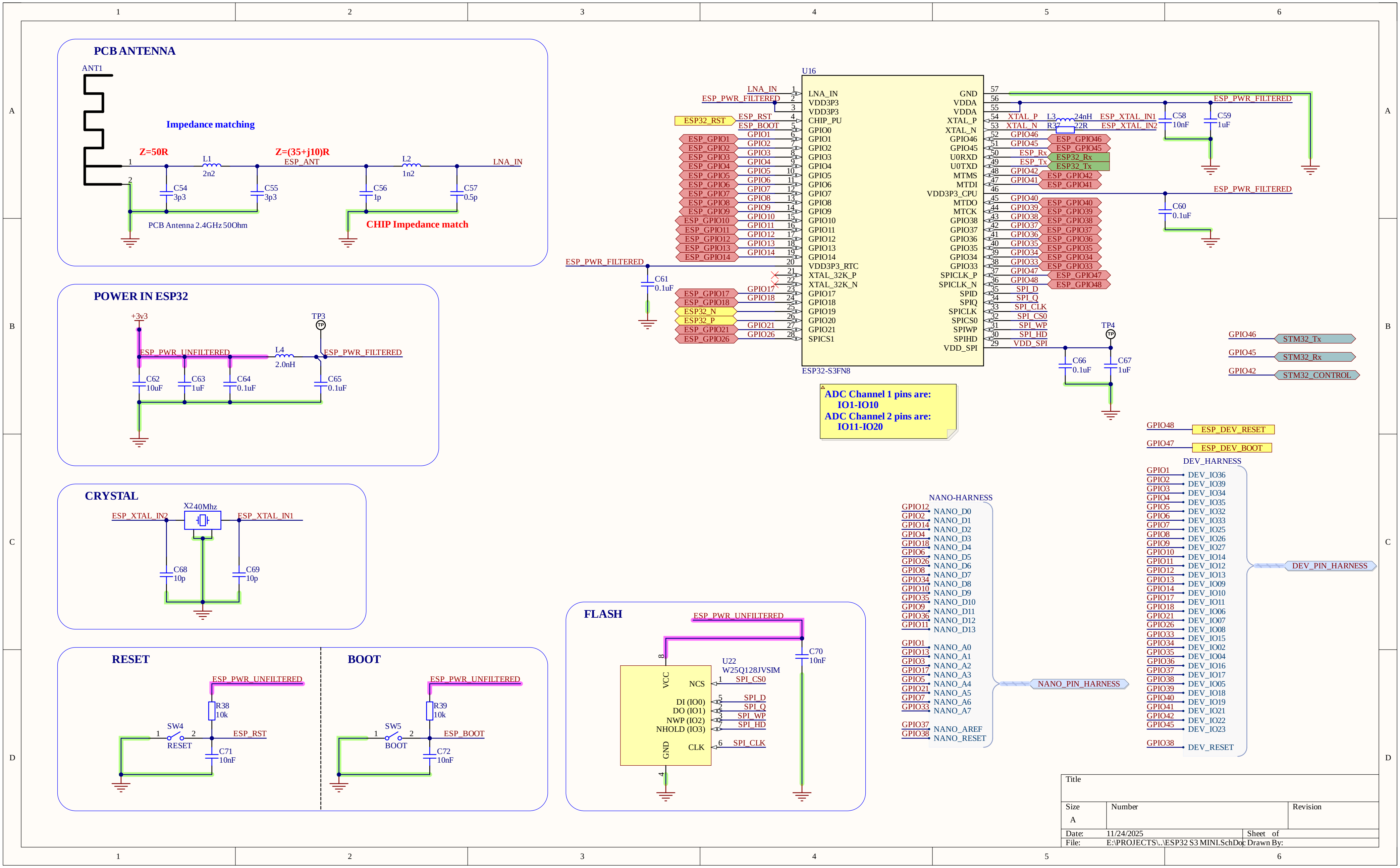
PCB ANTENNA
ANT1
Impedance matching
Z=50R
Z=(35+j10)R
ESP_ANT
LNA_IN
L1 2n2
L2 1n2
C54 3p3
C55 3p3
C56 1p
C57 0.5p
PCB Antenna 2.4GHz 50Ohm
CHIP Impedance match
POWER IN ESP32
+3v3
TP3
ESP_PWR_UNFILTERED
L4 2.0nH
ESP_PWR_FILTERED
C62 10uF
C63 1uF
C64 0.1uF
C65 0.1uF
CRYSTAL
X2 40Mhz
ESP_XTAL_IN2
ESP_XTAL_IN1
C68 10p
C69 10p
RESET
SW4 RESET
R38 10k
ESP_RST
C71 10nF
BOOT
SW5 BOOT
R39 10k
ESP_BOOT
C72 10nF
FLASH
U22 W25Q128JVSIM
VCC
NCS
DI (IO0)
DO (IO1)
NWP (IO2)
NHOLD (IO3)
GND
CLK
SPI_CS0
SPI_D
SPI_Q
SPI_WP
SPI_HD
SPI_CLK
C70 10nF
U16
ESP32-S3FN8
LNA_IN
VDD3P3
CHIP_PU
GPIO0
XTAL_32K_P
XTAL_32K_N
VDD3P3_RTC
GND
VDDA
XTAL_P
XTAL_N
U0RXD
U0TXD
MTMS
MTDI
VDD3P3_CPU
MTDO
MTCK
SPICLK_P
SPICLK_N
SPID
SPIQ
SPICLK
SPICS0
SPIWP
SPIHD
VDD_SPI
ESP32_RST
ESP32_N
ESP32_P
ESP32_Rx
ESP32_Tx
L3 24nH
R37 22R
C58 10nF
C59 1uF
C60 0.1uF
C61 0.1uF
TP4
C66 0.1uF
C67 1uF
ADC Channel 1 pins are:
IO1-IO10
ADC Channel 2 pins are:
IO11-IO20
STM32_Tx
STM32_Rx
STM32_CONTROL
ESP_DEV_RESET
ESP_DEV_BOOT
NANO-HARNESS
NANO_PIN_HARNESS
DEV_HARNESS
DEV_PIN_HARNESS
Title
Size A
Number
Revision
Date: 11/24/2025
Sheet of
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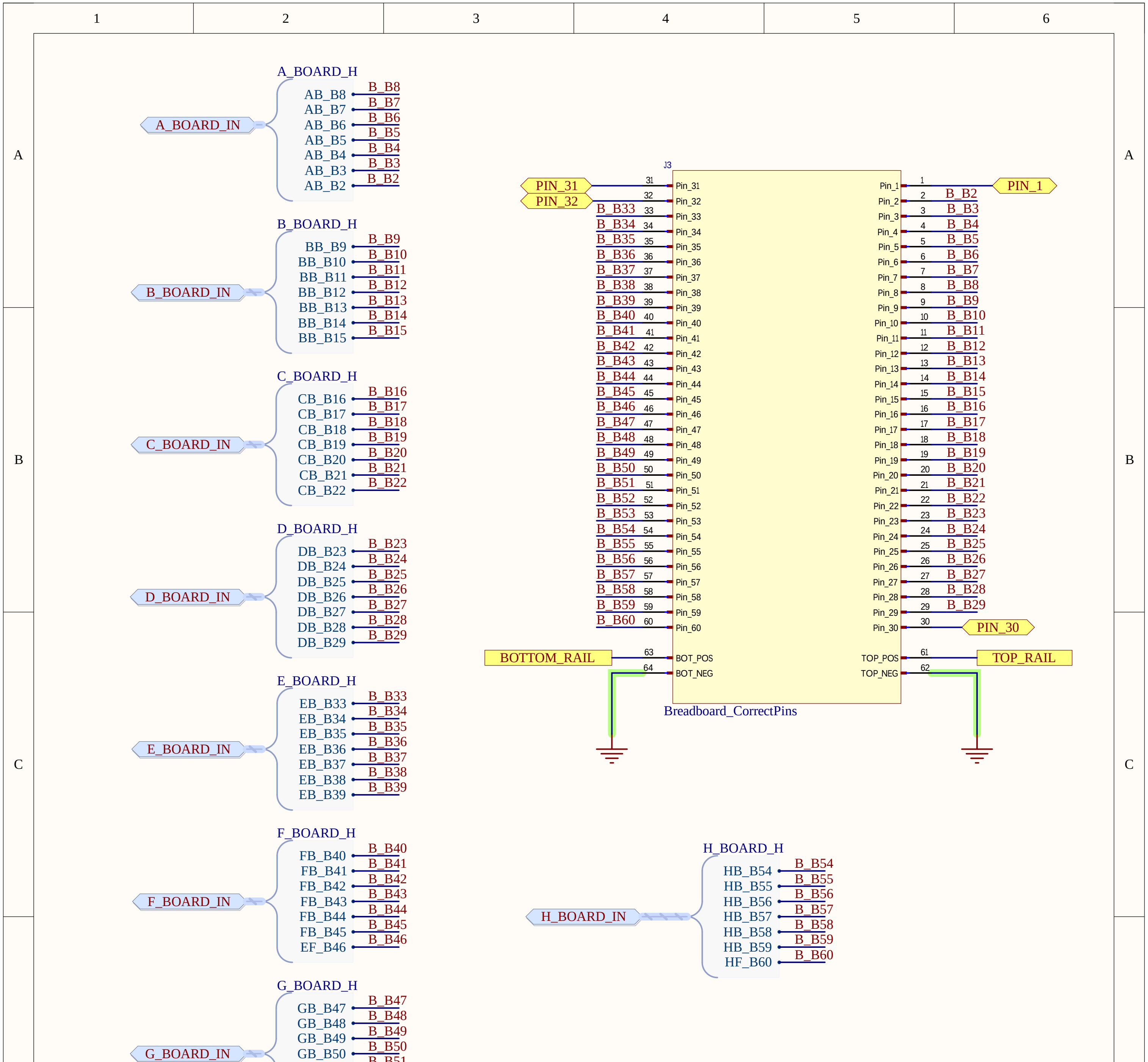

Title		
Size A	Number	Revision
Date: 11/24/2025		Sheet of
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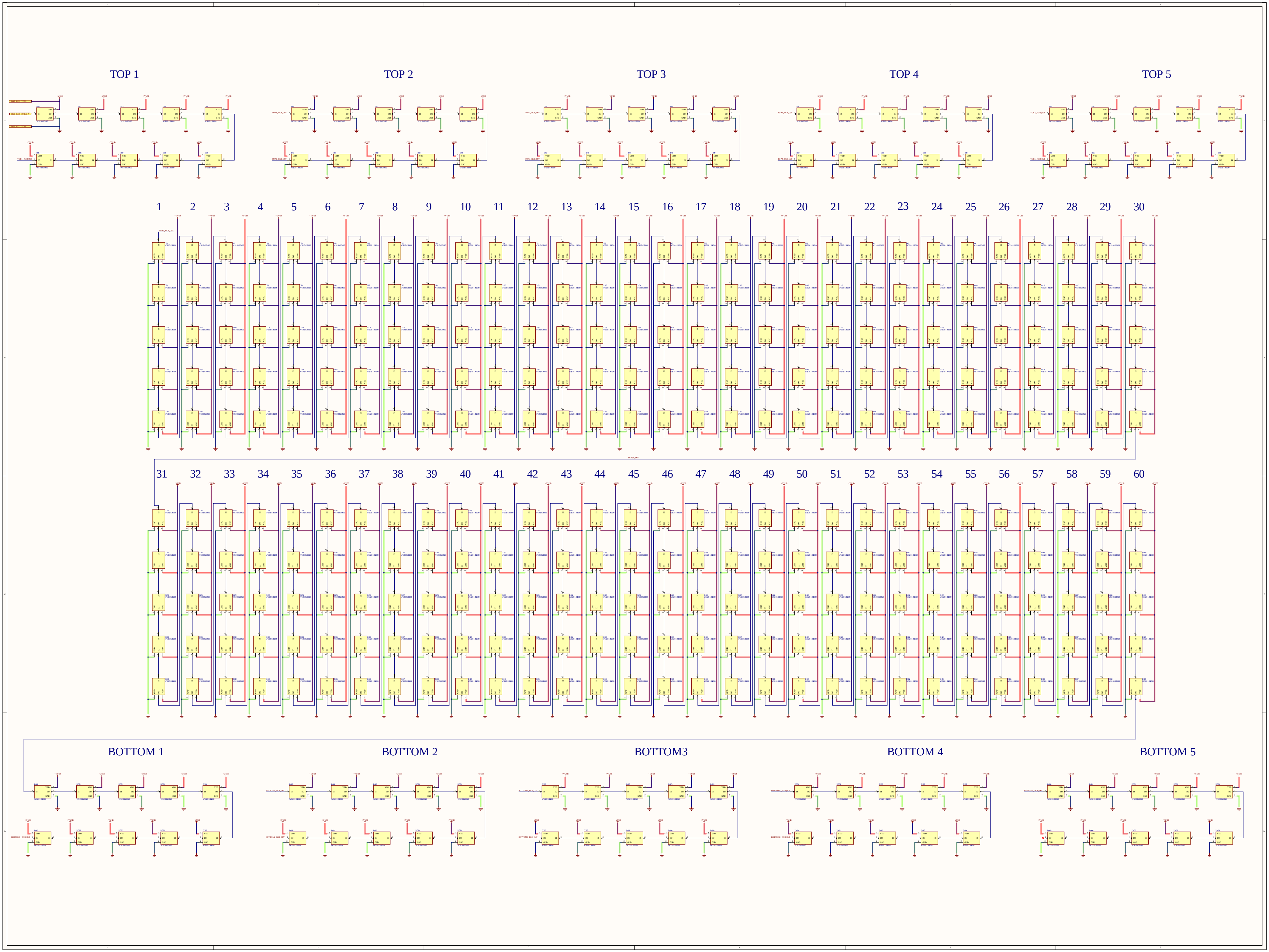
TOP 1
TOP 2
TOP 3
TOP 4
TOP 5
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
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43
44
45
46
47
48
49
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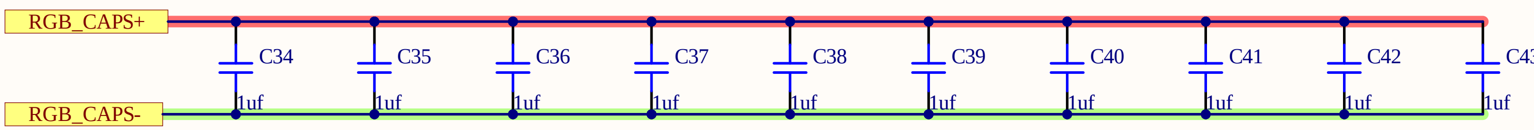
51
52
53
54
55
56
57
58
59
60
BOTTOM 1
BOTTOM 2
BOTTOM3
BOTTOM 4
BOTTOM 5



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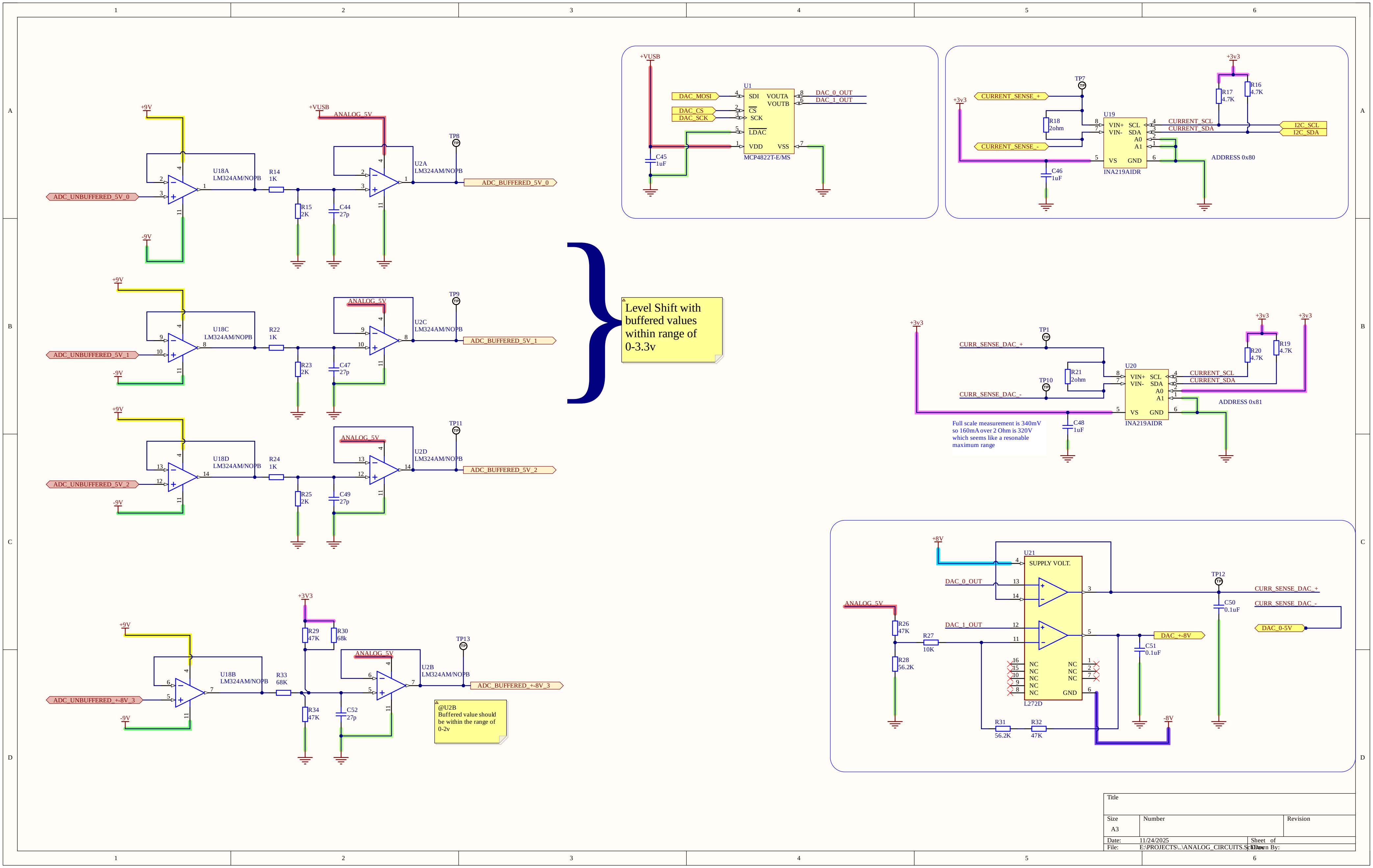
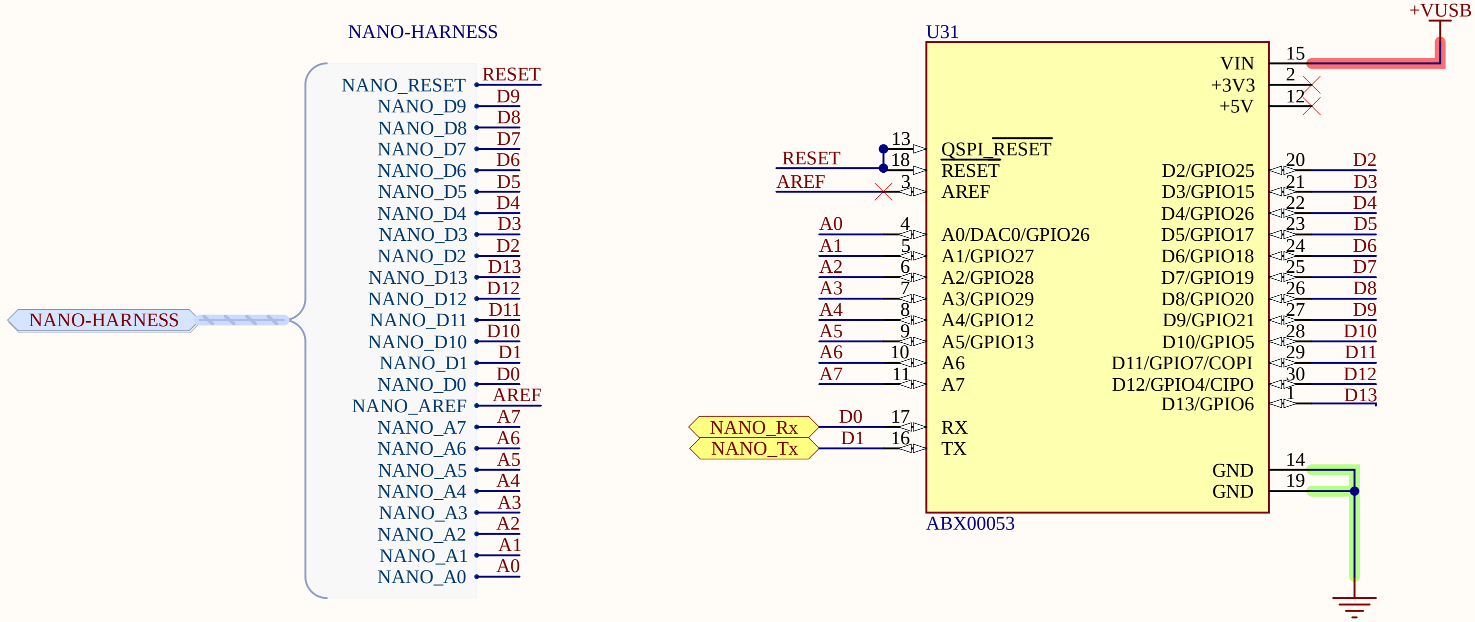
Size A	Number		Revision
Date:	11/24/2025	Sheet of	
File:	E:\PROJECTS\..\WS2812B_CAP_SET.SchDoc	Drawn By:	



ADC_UNBUFFERED_5V_0
U18A
LM324AM/NOPB
R14
1K
R15
2K
C44
27p
U2A
LM324AM/NOPB
TP8
ANALOG_5V
+VUSB
+9V
-9V
ADC_BUFFERED_5V_0
ADC_UNBUFFERED_5V_1
U18C
LM324AM/NOPB
R22
1K
R23
2K
C47
27p
U2C
LM324AM/NOPB
TP9
ADC_BUFFERED_5V_1
ADC_UNBUFFERED_5V_2
U18D
LM324AM/NOPB
R24
1K
R25
2K
C49
27p
U2D
LM324AM/NOPB
TP11
ADC_BUFFERED_5V_2
ADC_UNBUFFERED_+-8V_3
U18B
LM324AM/NOPB
R33
68K
+3V3
R29
47K
R30
68K
R34
47K
C52
27p
U2B
LM324AM/NOPB
TP13
ADC_BUFFERED_+-8V_3
@U2B
Buffered value should be within the range of 0-2v
Level Shift with buffered values within range of 0-3.3v
DAC_MOSI
DAC_CS
DAC_SCK
U1
SDI
CS
SCK
LDAC
VDD
VOUTA
VOUTB
VSS
MCP4822T-E/MS
DAC_0_OUT
DAC_1_OUT
C45
1uF
CURRENT_SENSE_+
CURRENT_SENSE_-
+3v3
TP7
R18
2ohm
U19
VIN+
VIN-
VS
SCL
SDA
A0
A1
GND
INA219AIDR
CURRENT_SCL
CURRENT_SDA
I2C_SCL
I2C_SDA
R17
4.7K
R16
4.7K
C46
1uF
ADDRESS 0x80
CURR_SENSE_DAC_+
CURR_SENSE_DAC_-
TP1
TP10
R21
2ohm
U20
INA219AIDR
R20
4.7K
R19
4.7K
C48
1uF
ADDRESS 0x81
Full scale measurement is 340mV so 160mA over 2 Ohm is 320V which seems like a resonable maximum range
+8V
U21
SUPPLY VOLT.
DAC_0_OUT
DAC_1_OUT
R26
47K
R27
10K
R28
56.2K
R31
56.2K
R32
47K
NC
GND
L272D
-8V
C51
0.1uF
DAC_+-8V
TP12
C50
0.1uF
CURR_SENSE_DAC_+
CURR_SENSE_DAC_-
DAC_0-5V
Title
Size
A3
Number
Revision
Date:
11/24/2025
Sheet of
File:
E:\PROJECTS\..\ANALOG_CIRCUITS.SchDoc
Drawn By:



Title		
Size: A	Number	Revision
Date: 11/24/2025		Sheet of
File: E:\PROJECTS\..\NANO_HEADER.SchDoc		Drawn By:

TYPICAL APPLICATION

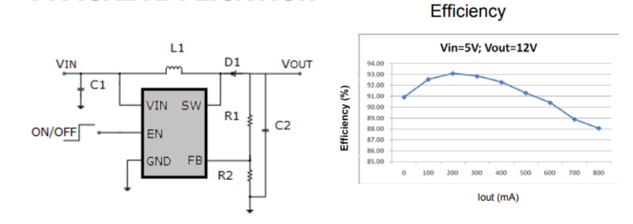
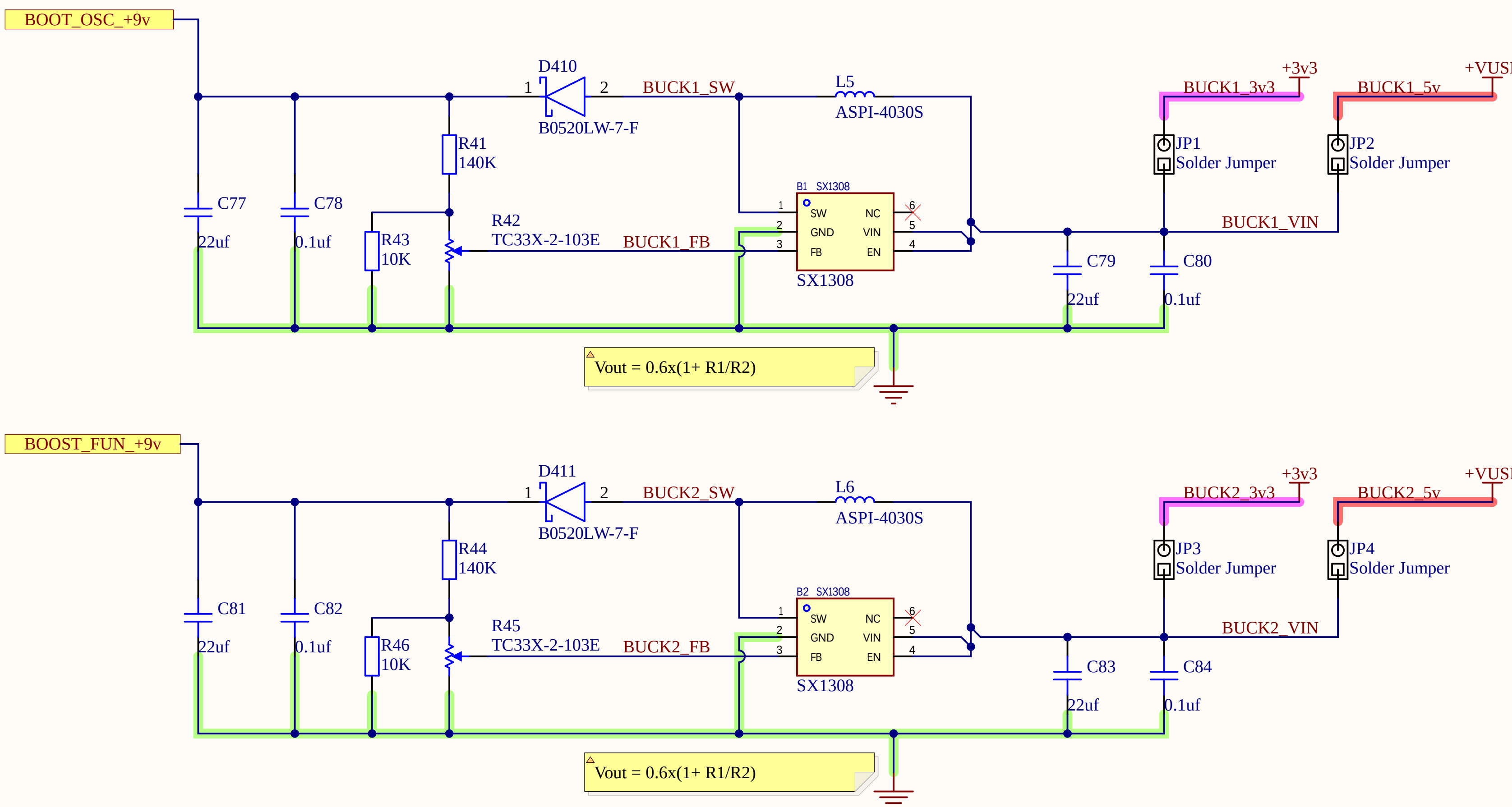


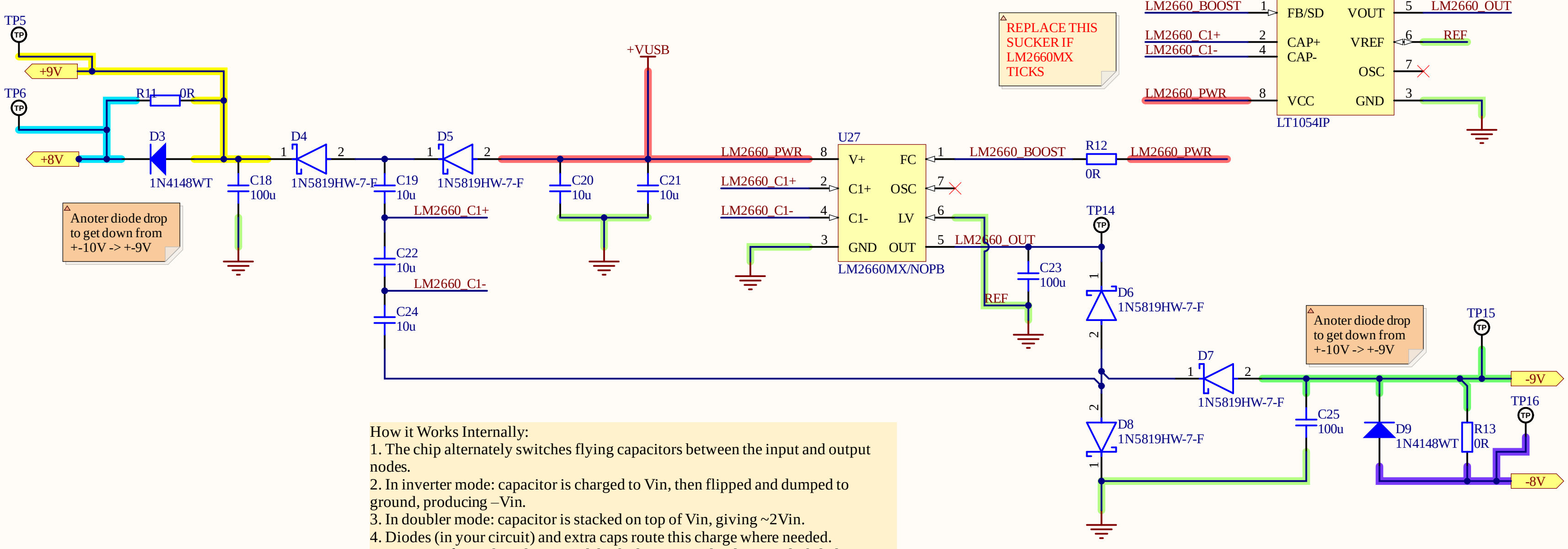
Figure 1. Basic Application Circuit

Figure 2. Efficiency Curve



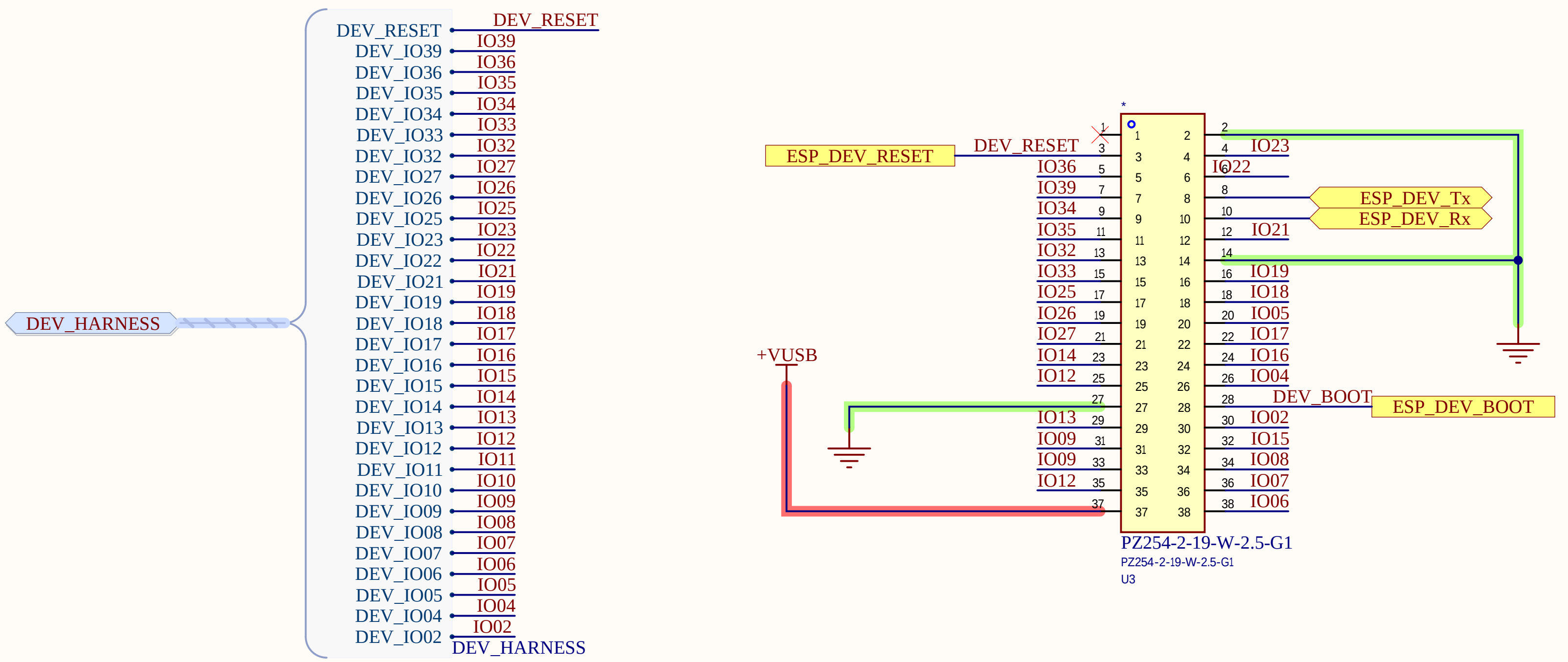
Title		
Size: A	Number	Revision
Date: 11/24/2025	Sheet of	
File: E:\PROJECTS\..\9v_BOOST_CONVERTERS.SchDoc	Drawn By:	

+-9V SUPPLY (VOLTAGE DOUBLER/ INVERTER)

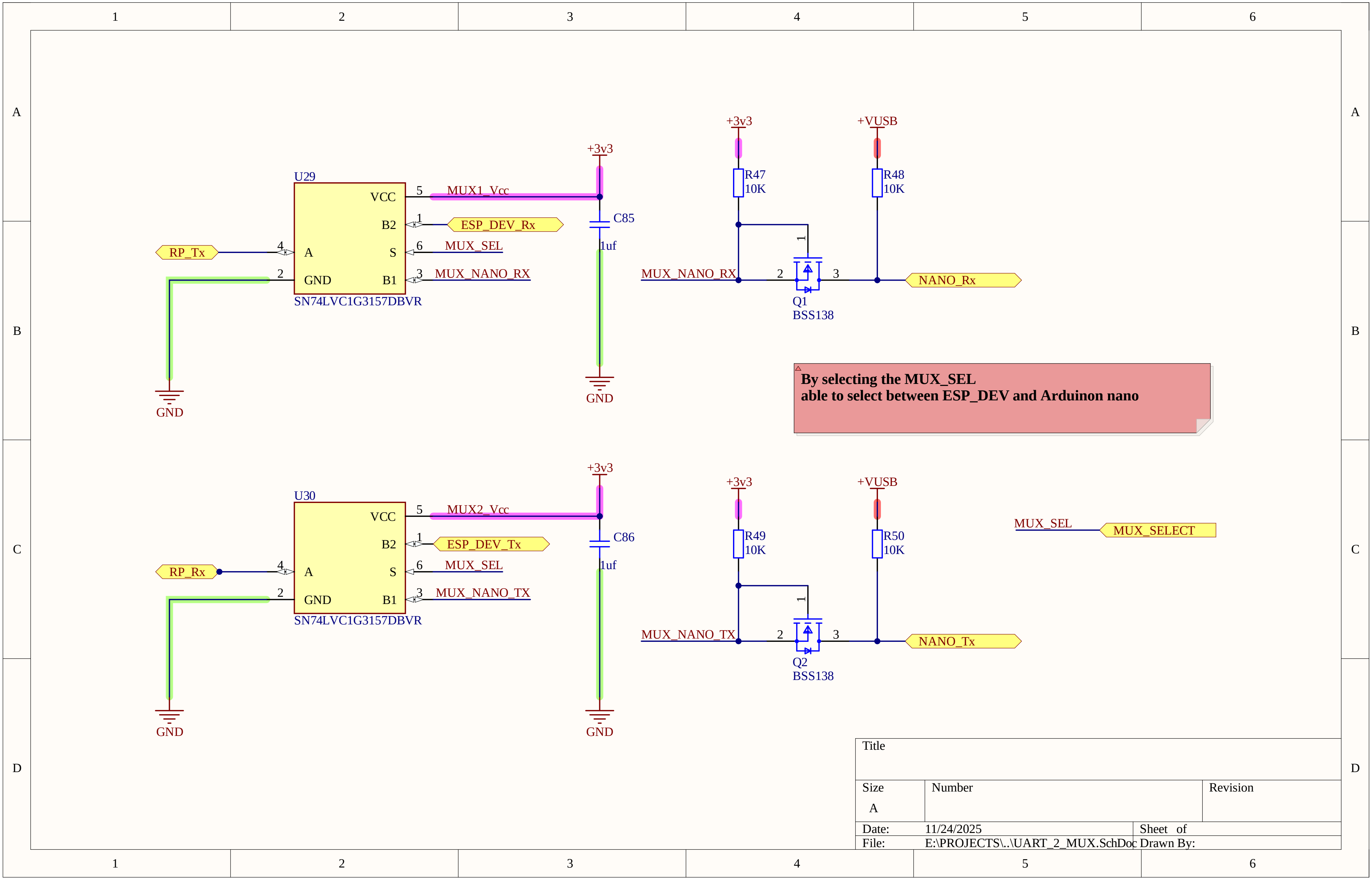


How it Works Internally:
1. The chip alternately switches flying capacitors between the input and output nodes.
2. In inverter mode: capacitor is charged to Vin, then flipped and dumped to ground, producing –Vin.
3. In doubler mode: capacitor is stacked on top of Vin, giving ~2Vin.
4. Diodes (in your circuit) and extra caps route this charge where needed.
5. Because of switching losses and diode drops, actual voltage is slightly less (hence the extra Schottky diodes to regulate to ±9V).

Title		
Size: A	Number	Revision
Date: 11/24/2025		Sheet of
File: E:\PROJECTS\..\+-9V Supply.SchDoc		Drawn By:



Title		
Size A	Number	Revision
Date: 11/24/2025		Sheet of
File: E:\PROJECTS\..\ESP32 DEV.SchDoc		Drawn By:


U29
SN74LVC1G3157DBVR
VCC
B2
A
S
GND
B1
MUX1_Vcc
ESP_DEV_Rx
MUX_SEL
MUX_NANO_RX
RP_Tx
GND
+3v3
C85
1uf
GND
+3v3
+VUSB
R47
10K
R48
10K
MUX_NANO_RX
Q1
BSS138
NANO_Rx
By selecting the MUX_SEL
able to select between ESP_DEV and Arduinon nano
U30
SN74LVC1G3157DBVR
MUX2_Vcc
ESP_DEV_Tx
MUX_SEL
MUX_NANO_TX
RP_Rx
GND
+3v3
C86
1uf
GND
+3v3
+VUSB
R49
10K
R50
10K
MUX_NANO_TX
Q2
BSS138
NANO_Tx
MUX_SEL
MUX_SELECT
Title
Size
A
Number
Revision
Date: 11/24/2025
Sheet of
File: E:\PROJECTS\..\UART_2_MUX.SchDoc
Drawn By: